
Comparison Navigation for Large Language Models and Humans

Márk István Hegedüs (AU777578), Kris Kraack (AU802637), Ruiqi Zhang (AU779759)

Abstract

Large language models (LLMs) excel at knowledge retrieval, but their ability to tackle structured navigation tasks like Wikipedia page navigation remains unclear. We evaluate LLMs under four paradigms (prompting strategies), including a novel strategy integrating external knowledge from knowledge graph developed in this study. Benchmarking on Wikispeedia dataset, we compare LLM and human navigation by success rate and other metrics. Our studies reveal that structured reasoning with these methods enables LLMs to achieve human-like or superior navigation performance to different extent. Stepwise analyses of semantic similarity and information gain show that LLMs initially explore broadly before converging to the target page, and specifically, the ToT strategy ensures steps remain semantically relevant and converges decisively on the appropriate category.

1 Introduction

Navigating complex information networks like Wikipedia requires multi-step reasoning and strategic planning. This project investigates how large language models reason through navigation tasks using the Wikispeedia dataset [West et al., 2009], which contains human navigation paths from a start to a destination Wikipedia article. Using human paths, we explore how model reasoning aligns or diverges from human behaviour. The goal is to uncover systematic reasoning tendencies in models and provide a grounded understanding of their strengths and limitations of structured, multi-step decision tasks compared to humans, identifying where and how model reasoning diverges from human navigation strategies.

Regarding practical usage, our work is related to Netflix movie recommendation, where the system suggests the next best title by combining local context with global structure (e.g.: genres, popular hubs, and common transition patterns). While this example doesn't have a specific destination item, it is still a form of navigation through a network by semantically aligned choices. With this project, we aim to answer the following question: Do LLMs show similar semantic navigation patterns to those of human players, and how do these patterns vary across prompting strategies and model sizes?

The remainder of this report is structured as follows: section 2 reviews prior studies on the Wikispeedia game navigation for LLMs and explains how our approach differs. section 3 describes the dataset, methods, and evaluation metrics used. section 4 presents the human versus LLMs comparison, and section 5 concludes our findings.

2 Related Work

2.1 Semantic and Strategic Analysis of Human Navigation

Wikipedia datasets, such as Wikispeedia, are naturally great resources for investigating human users' cognitive patterns in terms of finding their way in a growing sea of information. [Rodi et al., 2017] reveal the existence of clear and regular patterns in the users' navigation behaviours such as very first page navigated by human players being more abstract and of high level. [Lamprecht et al., 2017]

summarize the common human navigation strategy by a two stages process of “zoom-out,home-in”. They also discuss how the structure of Wikipedia articles influenced user navigation. [Tettoni and Primbault, 2024] focus on the ‘rabbit hole’ phenomena in human navigation, conducting a large-scale quantitative characterisation of the navigation traces of readers who fell into a wiki rabbit hole.

2.2 LLM Navigation on Wikipedia Datasets

Before the rise of LLM-based approaches, Graph-based neural architectures such as Relational Graph Convolutional Networks have been evaluated on multi-hop reasoning tasks[Staliūnaitė et al., 2022]. Large Language Models are supposed to be able to internalize the structural and relational properties of graph-based resources. Recent work has shown their potential in reasoning, pathfinding, and multi-hop navigation on graph-structured data. Games based on Wikispedia data, like WikiGame and Wikispedia, have long been a benchmark for human semantic navigation and have been recently used for evaluating LLMs’ performance on this task. [Margiotta et al., 2025] present a large-scale comparison of human and LLM navigation on the WikiGame, evaluating LLMs’ performance across a spectrum of diverse conditions.

2.3 Our Contributions

In our study, we provide a systematic comparison between humans and LLMs with modern prompting strategies. In addition to the commonly used strategies, we also develop a novel method enhanced by external graph and semantic knowledge. Furthermore, we explore and discuss the semantic progress and categorical shift of human and LLM-generated paths, enabling us to probe the nature of the reasoning of different agents.

3 Methodology

3.1 Experimental Paradigms

We evaluate LLMs under four progressively informative settings:

Blind Navigation (Direct Path Prediction). In this setting, the model is asked to directly output a valid path of a given pair of start and destination as short as possible, with no set of valid outgoing links from intermedium nodes provided. Errors like “hallucinated” nodes (page titles that do not exist in Wikipedia) or “hallucinated” links (hyperlinks between two pages that do not exist) may occur. The exact prompt is shown in Appendix 6.1.

Link-Aware Navigation with Chain-of-Thought Reasoning. Under this mode, at each step, the model is also provided with the set of outgoing links from the current node and is asked to choose the next Wikipedia page to traverse. Besides, the Chain-of-Thought (CoT) strategy is applied by requiring the model to articulate the reasoning behind each navigational step. The complete prompt is given in Appendix 6.2.

Link-Aware Navigation with External Information from Knowledge Graph. While LLMs are highly effective at learning probabilistic language patterns from large corpora, they still struggle with generating knowledge-grounded content. Recent studies such as KG-based Random-Walk Reasoning[Kim et al., 2024] and KG-RAR[Wu et al., 2025] have demonstrated the potential of integrating knowledge graph (KG) to LLM reasoning. Here, we propose a novel prompt strategy combining global graph and semantic information.

This model extends the naïve CoT by retrieving three supporting information of interest for each potential next page: The semantic similarity to the target page, the centrality score and the degree in the page hypergraph. See Appendix 6.3 for the full prompt.

Link-Aware Navigation with Tree-of-Thought Reasoning. In the Tree-of-thought (ToT) setting, the model operates in the same setup as link-aware with CoT, but is encouraged to explore multiple reasoning branches before committing to an action. At each step, the model considers several candidate next pages in parallel, while keeping track of the total score for all the current candidate paths. Pruning is applied to reduce time complexity. The prompt is identical to that of Link-Aware CoT.

3.2 Evaluation Metrics

To evaluate the navigation and reasoning capabilities of LLMs in relative to humans, under different prompting strategies, we compute the following metrics:

Success Rate. The success rate is defined by the proportion of LLM-generated paths in which the agent (human or LLM of a specific prompting strategy) successfully reach the target articles from the start page via a valid sequence of links. Invalid Links between two pages are checked.

Mean Path Length. Navigation efficiency is evaluated by the lengths of successful paths. For all successful paths, we compute the average number of steps to reach to goal across different agents. A shorter average path length may suggest the strategy used is more informative.

Computational Workload The computation workload of a model is the sum of the input and the output tokens. This measure reflects the inference cost for a given model and, when considered alongside task performance, enables an assessment of the model’s overall efficiency.

3.3 Experimental Evaluation

We start with collecting the publicly available metadata of Wikispeedia game from Stanford Network Analysis Platform (<https://snap.stanford.edu>), including finished paths, unfinished paths, articles and links. To make our analysis representative, we stratify our dataset of human navigation links by difficulty. For every unique start and destination pair, we annotate them based on their success rate of human players, specifically, Medium (50% success rate < 75%) and Hard (25% success rate < 50%). The category with a success rate of more than 75% is excluded as it only contains several pages, and those paths with less than 25% success rate or even impossible are not considered for potentially taking too much inference time. For the two categories that remained, we select the 30 most-played games, and execute the LLM generation twice, producing 120 LLM-generated paths corresponding to the human data.

For the experiments, the LLM API we choose are openai-gpt-oss-20b and openai-gpt-oss-120b, so we can compare how the scale of the model effects the performance, unbiasedly under the same GPT model family. Experiments were executed on a DigitalOcean cloud instance, which served as the client environment for issuing API calls to OpenAI-hosted language models. API calls were executed with fixed temperature($T = 0$) to ensure reproducibility.

4 Results

4.1 Comparing Human and Model Success

Figure 1 reports the success rates of each model and human participants across the different experiment settings. For both humans and LLMs, the success rates increase as task difficulty increases. Overall, all LLM-based models outperform human players. Larger LLM model (gpt-oss-120b) achieves higher performance in general as difficulty rises, whereas smaller models can approach or match the larger models on medium-difficulty tasks. Success rates also increase with the use of more computationally intensive reasoning strategies, especially ToT. Notably, CoT augmented with external knowledge benefits from increased model parameters, whereas it underperforms the baseline CoT on the smaller model. Appendix Figure 8 details the token usage as a measure of computational workload, enabling comparison across prompting strategies. We further assess path quality from another perspective by calculating average path lengths, shown in Appendix Figure 9.

4.2 Distribution of Path Lengths

Figure 2 illustrates the distribution of navigation path lengths for human and LLM players. Examining the distributions and associated whiskers reveals the variability in how the agents traverse the search space. The maximum path length for humans is constrained to 11, as going beyond this usually causes loops or hallucinations for the LLMs. From the distribution and interquartile range that humans generally have the longest paths, and that CoT paths with and without knowledge graph information have similar distribution, including shortest path length of two. ToT path lengths have very few outliers and rarely exceed seven. The difference in navigation path lengths makes direct comparison difficult. To address this, subsequent analyses (except the Sankey diagrams) in sections subsection 4.3

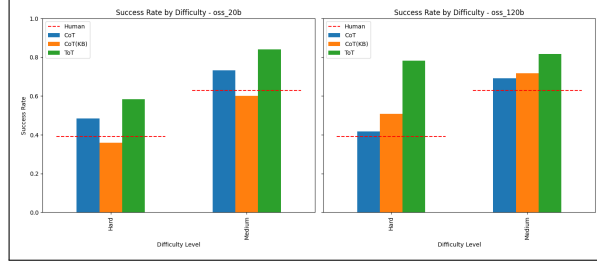


Figure 1: Success rates achieved by humans and different LLMs.

and subsection 4.4 use normalised path length, which is defined as progress along a path expressed as a percentage of the total path length. The normalisation enables fair comparison across agents with differing absolute path lengths and allows stepwise metrics to be aligned on a common scale.

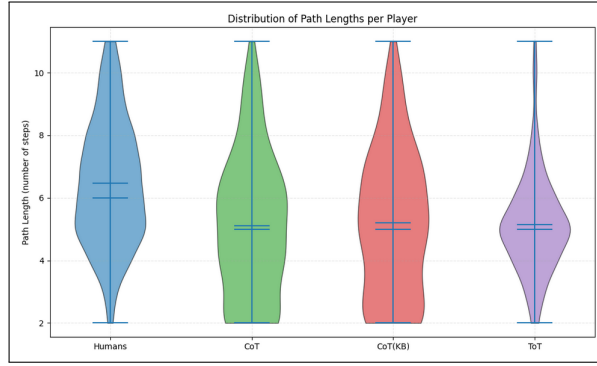


Figure 2: Violin plot showing navigation path lengths distributions comparison.

4.3 Semantical Similarity

Figure 3 shows the stepwise mean semantic similarity between consecutive article nodes along normalised navigation paths. For all players except Chain-of-Thought (CoT), semantic similarity decreases initially before increasing, indicating an exploratory phase. For Tree-of-Thought (ToT), the similarity value remains relatively constant within the middle range of the normalised progress. From around 70–75% of the normalised progress, all agents exhibit a significant increase in semantic similarity, followed by a plateau as they converge on the correct target article node. This pattern is consistent with previously reported search dynamics, which have been described as zooming out before zooming in: agents initially explore broadly, and then increasingly zoom in as they approach the destination [West et al., 2009].

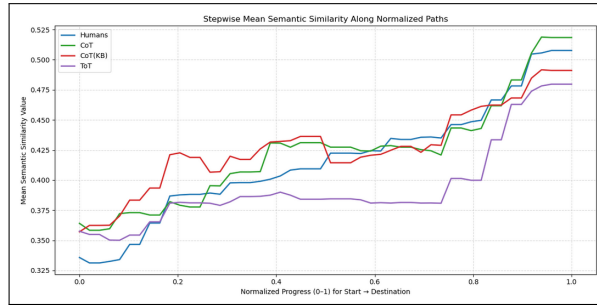


Figure 3: Step-wise mean similarity gain along normalized paths.

Further supporting the zooming behaviour of all the LLM agents, Figure 4 reports the stepwise mean information gain along normalised paths. The information gain trajectories mirror the semantic

similarity patterns: the early steps are characterised by lower or declining gains, which are then followed by a pronounced increase in the later stages. This suggests that meaningful, target-relevant information is primarily accumulated in the latter part of the path.

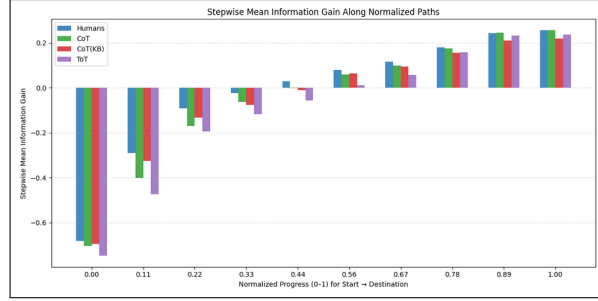


Figure 4: Step-wise mean information gain along normalized paths.

4.4 Categorical Shifts

Figure 5 illustrates the stepwise mean categorical shifts made by human players and large language models (LLMs). Both humans and most LLMs begin to focus increasingly on the destination category around the halfway point of the normalised progress, with categorical shifts gradually decreasing thereafter until the destination category is reached. In contrast, Tree-of-Thought (ToT) exhibits delayed category focusing, tending to spend more time on category exploration until around 80% of normalised progress has been reached. After this point, there is a sharp drop in categorical shifts, indicating rapid convergence towards the destination category.

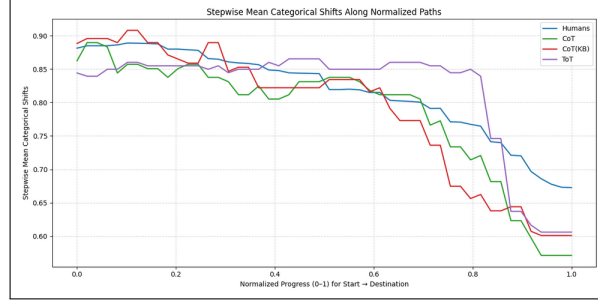
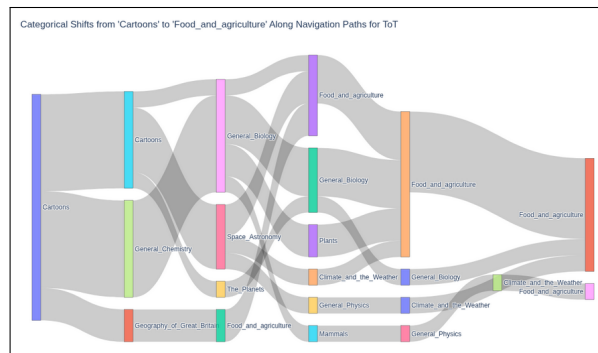
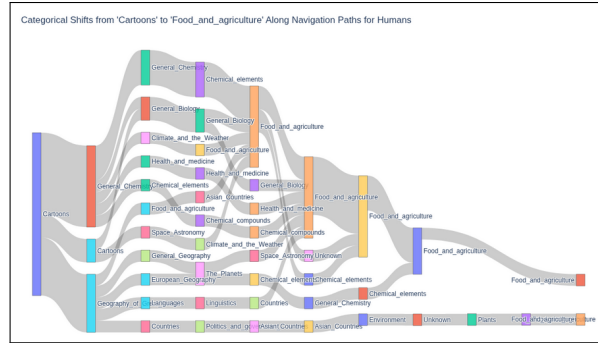


Figure 5: Step-wise mean categorical shifts along normalized paths.

This pattern is further illustrated when examining a specific start and destination article category is examined using Sankey diagrams (e.g. cartoons → food and agriculture). Plotted using non-normalised stepwise lengths, these diagrams demonstrate how category shifts unfold along the actual path. For example, in figure 5, human players, we use a sample size corresponding to the generated paths by the ToT model and the same constraint from the prompting strategy. Figure 6 illustrates that, by step four, some human players have finished the game. After step four, a progressively higher proportion of human players reach the destination category and finish the game. This behaviour corresponds to the gradual decrease in normalised progress after the halfway point.

Figure 7 shows the Sankey diagram for the ToT-generated paths. Examining the plot, we can see that once the ToT has focused on the destination category around step 3, it either finishes there or is highly likely to stay within or move onto the destination category. At step 4, the destination category accounts for over 70% of the distribution, which corresponds to the sharp decrease observed for the ToT at around 80% of the normalised progress. Another interesting observation is that, after step 3, if the ToT agent reaches the destination category, it either finishes or stays within it; it never exits the category. In contrast, human traversals are more widely scattered across categories during navigation, indicating less efficient category-level focusing.



5 Conclusion

The results demonstrate differences in navigation strategies between human players and large language models, particularly Tree-of-Thought (ToT). Analysis of path lengths shows that humans generally take longer and more variable routes, whereas ToT produces shorter paths with few outliers, indicating more consistent and efficient navigation. Normalising path length allows for a meaningful comparison of stepwise behaviours and reveals systematic patterns that would otherwise be obscured by differences in absolute traversal length.

We find that LLMs achieve higher navigation success rates than human players across our settings. Performance is strongest for the larger model (gpt-oss-120b), particularly on harder instances, while the smaller model remains competitive on medium-difficulty tasks. Success further improves with more compute-intensive reasoning strategies such as ToT.

Across normalised paths, both semantic similarity and information gain exhibit a shared structure across agents, characterised by an initial exploratory phase, followed by focused convergence towards the target. This zooming-out-and-in behaviour is evident for both human and LLM players, suggesting a common high-level search dynamic. However, categorical shift analysis reveals a key distinction: while humans and most LLMs gradually reduce category exploration after the midpoint of the path, ToT maintains category transitions until later stages before rapidly converging. Sankey analyses confirm that, once ToT enters the destination category, it remains there until completion; in contrast, humans continue to transition across multiple categories. Together, these findings suggest that ToT does not achieve efficient navigation through early commitment, but rather through constrained exploration combined with decisive category-level commitment rather than broad or unfocused exploration.

References

- Robert West, Joelle Pineau, and Doina Precup. Wikispeedia: An online game for inferring semantic distances between concepts. In *Proceedings of the Twenty-First International Joint Conference on Artificial Intelligence*, pages 1598–1603, 2009.
- Giovanna Chiara Rodi, Vittorio Loreto, and Francesca Tria. Search strategies of wikipedia readers. *PloS one*, 12(2):e0170746, 2017.
- Daniel Lamprecht, Kristina Lerman, Denis Helic, and Markus Strohmaier. How the structure of wikipedia articles influences user navigation. *New Review of Hypermedia and Multimedia*, 23(1): 29–50, 2017.
- Anne-Laure Tettoni and Simon Dumas Primbault. Going down the rabbit hole: Characterizing the long tail of wikipedia reading sessions. In *Computational Humanities Research*, pages 1324–1330, 2024.
- Ieva Staliūnaitė, Philip John Gorinski, and Ignacio Iacobacci. Relational graph convolutional neural networks for multihop reasoning: A comparative study. *arXiv preprint arXiv:2210.06418*, 2022.
- Daniele Margiotta, Danilo Croce, and Roberto Basili. Evaluating large language models on wikipedia graph navigation: Insights from the wikigame. 2025.
- Yejin Kim, Eojin Kang, Juae Kim, and H Howie Huang. Causal reasoning in large language models: A knowledge graph approach. *arXiv preprint arXiv:2410.11588*, 2024.
- Wenjie Wu, Yongcheng Jing, Yingjie Wang, Wenbin Hu, and Dacheng Tao. Graph-augmented reasoning: Evolving step-by-step knowledge graph retrieval for llm reasoning. *arXiv preprint arXiv:2503.01642*, 2025.

6 Appendix

6.1 Prompt:Blind Navigation

Prompt: Blind

You are playing the Wikispeedia game.

Start page: "Brain"

Goal page: "Telephone"

Think step-by-step and suggest a possible path of Wikipedia pages that could help reach the goal.

- Each page in the path must be a valid Wikipedia page, and there must be a link between each consecutive page in wikipedia.
- At each step, explain briefly why you're choosing that link.
- The path should start with the start page and end with the goal page.

Once you've reached the destination, write the full path.

Return your answer strictly following this JSON schema:

```
{'properties': {'pages': {'items': {'type': 'string'}, 'title': 'Pages', 'type': 'array'}}, 'required': ['pages'], 'title': 'Path', 'type': 'object'}
```

6.2 Prompt:Link-Aware Navigation with Chain-of-Thought

Link-Aware Navigation with Chain-of-Thought

You are playing the Wikispeedia game.
Current page: "Brain"
Goal page: "Telephone"

Think step-by-step and suggest exactly 3 possible next Wikipedia pages that could help reach the goal.

- Only choose from this list of valid outgoing links: ['Action_potential', 'Algorithm', "Alzheimer's_disease", 'Animal', 'Archaeology', 'Aristotle', 'Arthropod', 'Bacteria', 'Bee', 'Biology', 'Cameroon', 'Cattle', 'Cell_(biology)', 'Cerebellum', 'Charles_Darwin', 'Chordate', 'Computer_science', 'Crustacean', 'Down_syndrome', 'Edema', 'Emotion', 'Europe', 'Eye', 'Fish', 'Gorilla', 'Herodotus', "Huntington's_disease", 'Insect', 'Invertebrate', 'Kuru_(disease)', 'Lyme_disease', 'Mammal', 'Mathematics', 'Matter', 'Medicine', 'Meningitis', 'Mexico', 'Mind', 'Multiple_sclerosis', 'Papua_New_Guinea', 'Philosophy', 'Philosophy_of_mind', 'Prion', 'Protein', 'Psychology', 'Reptile', 'Schizophrenia', 'Sleep', 'Southern_United_States', 'Stroke', 'Tay-Sachs_disease', 'Tribe', 'Vertebrate', 'Virus']
- For each suggestion, include:
 - * next_page (must be exactly one of the valid_links)
 - * rating (1-10, higher = more promising)

Return your answer strictly following this JSON schema:

```
{'$defs': {'Candidate': {'properties': {'next_page': {'title': 'Next Page', 'type': 'string'}, 'rating': {'exclusiveMinimum': 0, 'maximum': 10, 'title': 'Rating', 'type': 'integer'}}}, 'required': ['next_page', 'rating'], 'title': 'Candidate', 'type': 'object'}}, 'properties': {'candidates': {'items': {'$ref': '#/$defs/Candidate'}, 'maxItems': 10, 'minItems': 3, 'title': 'Candidates', 'type': 'array'}}, 'required': ['candidates'], 'title': 'NavigationStep', 'type': 'object'}
```

6.3 Prompt:Link-Aware Navigation with External Information

Link-Aware Navigation with External Information

You are playing the Wikispeedia game.
Current page: "Brain"
Goal page: "Telephone"

Think step-by-step and suggest exactly 3 possible next Wikipedia pages that could help reach the goal.

- Only choose from this list of valid outgoing links: ['Action_potential', 'Algorithm', "Alzheimer's_disease", 'Animal', 'Archaeology', 'Aristotle', 'Arthropod', 'Bacteria', 'Bee', 'Biology', 'Cameroon', 'Cattle', 'Cell_(biology)', 'Cerebellum', 'Charles_Darwin', 'Chordate', 'Computer_science', 'Crustacean', 'Down_syndrome', 'Edema', 'Emotion', 'Europe', 'Eye', 'Fish', 'Gorilla', 'Herodotus', "Huntington's_disease", 'Insect', 'Invertebrate', 'Kuru_(disease)', 'Lyme_disease', 'Mammal', 'Mathematics', 'Matter', 'Medicine', 'Meningitis', 'Mexico', 'Mind', 'Multiple_sclerosis', 'Papua_New_Guinea', 'Philosophy', 'Philosophy_of_mind', 'Prion', 'Protein', 'Psychology', 'Reptile', 'Schizophrenia', 'Sleep', 'Southern_United_States', 'Stroke', 'Tay-Sachs_disease', 'Tribe', 'Vertebrate', 'Virus']

- For each suggestion, include:

- * next_page (must be exactly one of the valid_links)
- * rating (1-10, higher = more promising)

Here is some additional external knowledge that might help you make better decisions:

```
{'Chordate': {'similarity_to_gaol': 0.45015963912010193,
'centrality_diff_to_goal': 0.0013046535958283444,
'degree': 0.18753387533875338}, 'Europe': {'similarity_to_gaol':
0.28999996185302734, 'centrality_diff_to_goal': 0.006008957440599388,
'degree': 0.591869918699187}, 'Medicine': {'similarity_to_gaol':
0.3622124195098877, 'centrality_diff_to_goal':
0.0005500223419375004, 'degree': 0.07696476964769648},
```

(The following 10+ lines are omitted)

```
'Southern_United_States': {'similarity_to_gaol': 0.261041522026062,
'centrality_diff_to_goal': 0.00012289525205949603, 'degree':
0.055284552845528454}},
```

you can use this information to inform your suggestions.

Return your answer strictly following this JSON schema:

```
{'$defs': {'Candidate': {'properties': {'next_page': {'title':
'Next Page', 'type': 'string'}, 'rating': {'exclusiveMinimum': 0,
'maximum': 10, 'title': 'Rating', 'type': 'integer'}}, 'required':
['next_page', 'rating'], 'title': 'Candidate', 'type': 'object'}},
'candidates': {'items': {'$ref': '#/$defs/Candidate'},
'maxItems': 10, 'minItems': 3, 'title': 'Candidates', 'type': 'array'}},
'required': ['candidates'], 'title': 'NavigationStep', 'type': 'object'}
```

6.4 Computation Cost Analysis

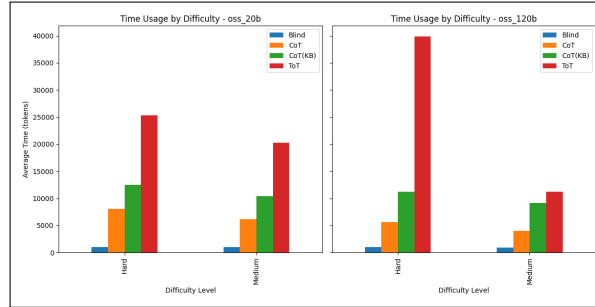


Figure 8: Computational cost measured by token usage.

6.5 Path Length Analysis

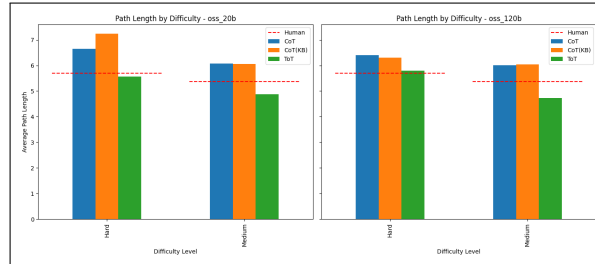


Figure 9: Average lengths of paths generated by humans and different LLMs.

6.6 Visualization of Human and LLM paths over Articles Embeddings

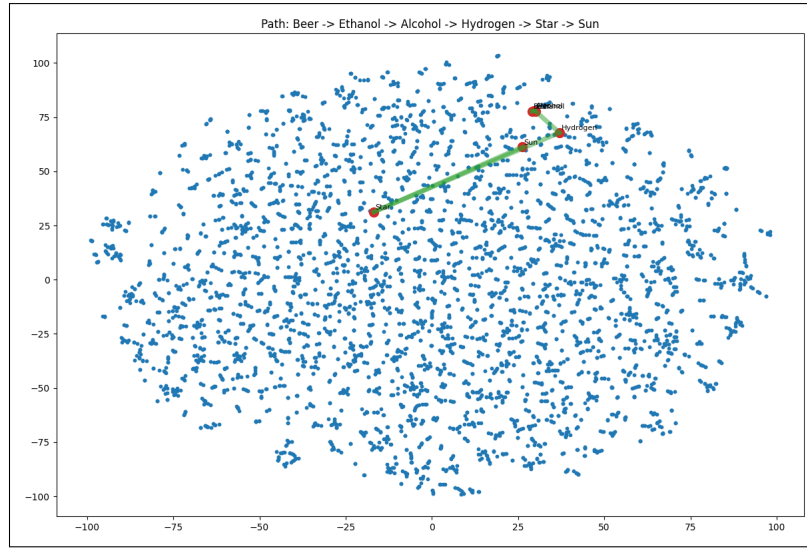


Figure 10: An example of paths from human players.

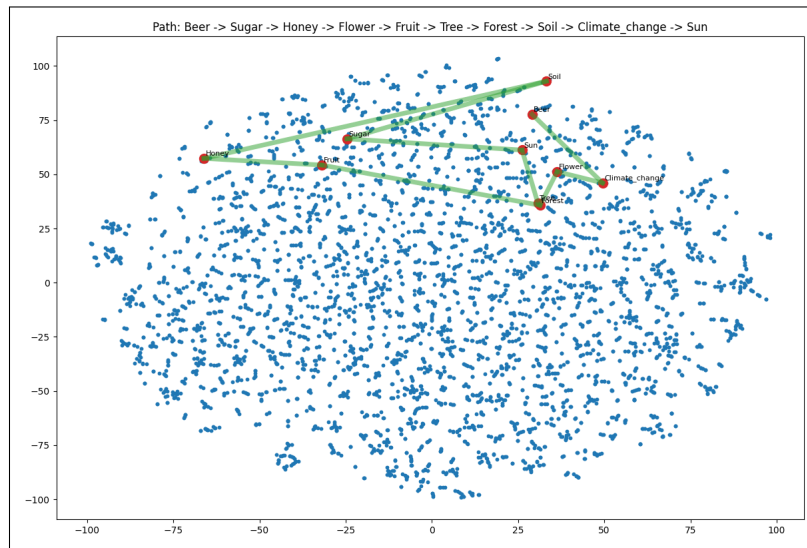


Figure 11: An example of paths generated by different LLMs.